

Non-stationary conditional extremes for BC streamflow: presenting the analysis, and what is inside the code

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This document exists for two jobs. **Job 1: present the analysis.** Part I walks the report in the order I would show it; for every figure it gives the axes, the theory behind the picture (with the derivation where one exists), the exact mechanism that produced it, what our own numbers say, and the follow-up questions that figure invites, with answers. **Job 2: prove the code is not a black box.** Part II opens each algorithm: what it computes, the pseudocode, why that is the right computation, which part is a published package and which part is ours, and how each is verified. Part III is a question-and-answer drill. Part IV is the fix list.

Pointers of the form “v8 § x , eq. (n), p. y ” give the section, equation and page of *my current understanding of the model*, version 8. Every number quoted is from the knit of `code_report_v4.Rmd` on `daily_sf_bf_rcp45_only_1945_2099_9avg.csv`: R 4.5.2, qgam 2.0.0, evgam 1.0.1, mgcv 1.9-3, texmex 2.4.9, extRemes 2.2-1, quantreg 6.1; axis 1969-01-01 to 2099-12-31, $n = 47,847$ days; conditioning pair BRIDG given NTHMB (Case 4), anchor BARRM given BARRS (Case 1).

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1 Status: what is ready to show, and what is not

Read this page before the meeting. The analysis has two halves, and they are in different states.

Stage	State	Basis
Data cleaning gate	present	0 duplicate rows, 0 missing dates, 0 NA, no non-positive baseflow; streamflow effectively tie-free (most repeated value occurs twice; 0.0% of days on values repeated ≥ 10 times)
Seasonal threshold	present	Threshold tracks the annual cycle in every window; trend slopes significant at all three series ($z = 11.8, 7.0, 17.7$)
Local 5% calibration	present, with the upgrade	Additive design misses the first era (SON 0.149, JJA 0.123); the tensor <code>te(doy, tt)</code> repairs it (SON 0.039, JJA 0.039)
GP tail	present	2236–2299 exceedances per series; scale trend significant everywhere ($p < 10^{-8}$)
PIT / Laplace gate	present	Histograms flat to within the noise the sample allows (Section 1.5 quantifies “flat”), sliced QQ on the line in both seasons and both decades, no point mass
Extremal index, return levels, machinery checks	present	extRemes $\equiv$ project to three decimals; Route A vs B agree to $\leq 1\%$
Dependence stage	do not present as a finding	The constrained optimiser did not move off its starting value: all three model sizes returned the identical objective 2698.6, the trend test returned exactly 0.00, $\hat{a}(t)$ came out exactly flat. Section 1.18 diagnoses it and gives the repair
Conditional functionals	report as provisional	They are computed from that stuck dependence fit, so the flat $P_{co} \approx 0.35$ is a property of the defect, not of the rivers

The honest sentence for the meeting: *the marginal stage passes every check you asked for; the dependence stage has a numerical defect I have located and can state precisely, and its results are on hold until it is fixed.* That is a stronger position than presenting a number that cannot be defended — and Section 1.18 shows the defect is already half-solved, because two independent pieces of evidence agree on what the answer should look like.

Part I — The walkthrough

1.1 Station 1: the data gate

What is shown. Printed counters (duplicates, gaps, missing values, non-positive baseflow), then three forensic instruments per series: the ten most repeated exact values with the months and years they fall in; runs of identical consecutive days; and the audit of days at or near the series minimum.

Why ties are fatal if present — the exact mechanism. The whole pipeline stands on the probability integral transform (Section 1.5.1), and that theorem requires a *continuous* distribution. Suppose instead some value x_0 occurs on many days, so $\Pr(X = x_0) = p > 0$. Then the transformed variable $U = F(X)$ inherits an atom: a mass of size p sits at the single point $F(x_0)$, no matter how well the model fits. On the histogram that is a spike; on the QQ plot it is a flat run of identical sample values ending in a jump; on the Laplace scatter it is a horizontal or vertical line of points. So the forensics are not tidiness — they are a precondition of the method, and each instrument targets one way mass can enter: the repeated-value table finds atoms wherever they occur; the run table finds *plateaus* (one value frozen across consecutive days, which is an instrument or post-processing signature, since rivers vary daily — and which also inflates serial dependence); the floor audit finds a hard minimum (detection limit or frozen gauge), the single most common atom in flow data.

How each table is computed. The repeated-value table is a frequency count of exact values, keeping the top ten, with each value’s day count, year range, number of distinct years, and the months it occurs in — the last two columns answer the professor’s question “which values are these, and is it only one year or many.” The run table applies run-length encoding to the daily series and keeps every run of ≥ 3 equal consecutive values with its start date, length and value, plus a month table of where runs live. The floor audit counts days exactly at the minimum and within 1% of it.

Our numbers. Streamflow, both stations: the most repeated value occurs exactly *twice* in 47,847 days; 0.0% of days sit on values repeated ten or more times; zero runs of three identical days; the minimum is carried by a single day. Baseflow: 3.0% of days (1447) carry a value repeated ≥ 10 times — the repeated values are spread over 10–14 different years each and concentrate in low-flow months — and there are exactly four short plateaus (lengths 3, 3, 3, 4; one in February, three in March). File-level: 0 duplicated (station, date) rows, 0 missing dates, 0 missing values, 0 days of $\text{bf} \leq 0$.

What to say. “The cleaning items are closed: no duplicated records, no gaps, no missing values, no non-positive baseflow, and the streamflow series carry no repeated values and no frozen stretches. The remaining tie mass is confined to baseflow — three percent of days, spread across years, in the low-flow range — which is a precision signature of the baseflow product, matters only for Structure B, and is documented rather than hidden.”

Question: “how do you know the winter data are not corrupted?” Three pieces of evidence, all printed: no winter value repeats (streamflow), no frozen stretches exist anywhere in streamflow, and the winter window figures show the smooth recessions hydrology produces rather than flat segments or discrete steps.

Question: “would you exclude the baseflow ties?” No — they are endemic (every era, not one bad year), so exclusion would delete a property of the data source rather than an error. The honest treatments are to note the granularity in the paper and, if Structure B tail diagnostics ever show an atom-induced artifact, to jitter within the reported precision as a sensitivity check.

1.2 Station 2: the seasonal moving threshold

What is shown. Twenty separate panels — both stations, streamflow and baseflow, five randomly drawn two-year windows each. Grey dots: daily values. Red: the working threshold $u(t; 0.95)$. Blue: the moving median $u(t; 0.5)$. Each title carries that window’s own share of days above red.

The principle: a threshold is a probability, not a level. The object is the conditional quantile curve defined by $\Pr\{X_t \leq u(t; \tau)\} = \tau$ at every date (v8 §2.1, eq. (3), p. 3). On a

snowmelt river a fixed horizontal level is a once-a-decade event in February and a routine event in July, so its exceedances would be “July days,” not “extreme days,” and every downstream stage would inherit that confusion. Fixing the probability and letting the level move makes “exceedance” mean the same thing — rarity $1 - \tau$ — on every date. That is Lee and Lee’s argument, and the local exceedance checks of Station 1.3 verify the property rather than assume it.

Why quantile regression finds a quantile — the derivation. The fitted curve minimises the pinball (check) loss $\rho_\tau(r) = \tau r$ for $r \geq 0$ and $(\tau - 1)r$ for $r < 0$, applied to residuals $r = x - u$. Differentiate the expected loss in u :

$$\frac{d}{du} \mathbb{E} \rho_\tau(X - u) = -\tau \Pr(X > u) + (1 - \tau) \Pr(X < u) = F(u) - \tau,$$

which is zero exactly when $F(u) = \tau$. So minimising the pinball loss *is* estimating the τ -quantile — not a heuristic, an identity. Everything else (splines, penalties, qgam) is machinery for letting u depend on t smoothly while this identity does the statistical work.

What the curve is made of. Two components in one formula (v8 §4.1–4.2, pp. 5–7). A *cyclic* day-of-year spline (`bs = "cc"`): basis functions periodic with period one year, built so the curve and its derivatives match across the 31 December / 1 January join — one seasonal shape, learned from all 131 years at once. Plus a slow *trend* spline in the running date: cubic bumps with knots spread over the century, each coefficient shaped by the several years under its own bump. The tensor upgrade of Station 1.3 replaces “seasonal shape + trend” by a smooth *surface* in (day-of-year, year), letting the shape itself evolve.

Why not fit each calendar day separately. Day-of-year alone has 366 covariate values and deletes the year — June 14 of 1970 and of 2095 would be forced to share one distribution, and change over the record (the object of study) could not appear. The running axis keeps the year but has no repeats; smoothness is what converts neighbouring days into information about a given date. The two-component basis takes the best of both: repetition across years for the cycle, neighbours along the axis for the trend.

The trend test behind the slope numbers (move one). The pinball loss is a loss, not a likelihood, so there is no likelihood-ratio test here. Instead: fit the rung with a linear trend term (harmonics included in *both* candidates, so the slope is trend, not season), then resample whole years with replacement 30 times, refit, and use the spread of the 30 slopes as the standard error. Whole years, because flood days cluster — resampling days would understate the error. Our numbers: NTHMB slope 61.8, SE 5.2, $z = 11.8$; BRIDG 7.12/1.02, $z = 7.0$; BRIDG baseflow 1.77/0.10, $z = 17.7$. All three high quantiles are rising, decisively.

What to say. “The threshold tracks the seasonal cycle within each year — these are the one-to-two-year windows you asked for — and each window’s own exceedance share is printed in its title. The trend in the high quantile is significant at every series.”

Question: “what should the share in a title be?” 0.05, with room for noise: a two-year window holds about 730 days, so even independent days give a standard deviation of $\sqrt{0.05 \times 0.95/730} \approx 0.008$, and clustering roughly doubles it — so anything in about 0.03–0.07 is consistent with perfect calibration, and a title far outside that range localises a miscalibrated stretch.

1.3 Station 3: the local 5% table — and why the tensor design wins

What is shown. The share of days above $u(t; 0.95)$ by season (rows) and 22-year era (columns) — 24 cells, target 0.05 in each — under the additive threshold; then the same table under the tensor

threshold.

Why this is the decisive check. Pooled calibration is guaranteed almost by construction (a quantile fit places 5% of *all* days above itself); the professor’s demand was “roughly 5% *locally*, not just overall,” i.e. that the table be flat. And the check is *out of model*: no fitting criterion optimises these cells, so improvement in them is evidence, not bookkeeping.

How much a cell may wobble. A cell holds roughly $22 \times 90 \approx 2000$ days. Independent days would give the share a standard deviation of $\sqrt{0.05 \times 0.95/2000} \approx 0.005$; clustering inflates it to roughly 0.01–0.015, so cells within about ± 0.03 of the target are noise. Cells at 0.123 and 0.149 are not noise — they are seven to ten standard deviations out.

Our numbers. Additive design, first era [1969,1991): SON 0.149, JJA 0.123, MAM 0.074, DJF 0.010 — the seasonal shape fitted as the 131-year average does not match the first decades, so the threshold sits too low in summer–autumn and too high in winter there; later eras are close to target. Tensor design `te(doy, tt)`, same first era: SON 0.039, JJA 0.039, MAM 0.043, DJF 0.012; all 24 cells lie in 0.012–0.053.

The concept behind the tensor. Additive $f(\text{doy}) + g(\text{tt})$ means *one* seasonal profile, the same in 1970 and 2095, shifted up or down by the trend. The tensor product builds a smooth surface from products of the two bases, so the freshet can arrive earlier, sharpen or flatten across the century — which on a warming snowmelt river is precisely the expected physics. The additive model is a special case of the tensor, so nothing is lost by the generalisation, and the smoothing penalty (chosen by marginal likelihood) shrinks back toward the simpler shape unless the data pay for the extra freedom.

What to say. “The additive threshold fails your local-5% test in the first era; letting the seasonal shape itself drift — a one-line change of formula — repairs every cell. I propose adopting the tensor design.”

1.4 Station 4: the GP tail

What is shown. A table (route, exceedance count, shape $\hat{\xi}$, trend test), the fitted scale $\sigma(t)$ averaged by era, and two boxplot panels: raw excesses by month, and excesses divided by the fitted $\sigma(t)$, by month.

Why generalized Pareto and nothing else. The Pickands–Balkema–de Haan theorem: for essentially any underlying distribution for which maxima have a limit law, the distribution of the *excess over a high threshold*, $X - u$ given $X > u$, converges as the threshold rises to a generalized Pareto distribution,

$$\Pr\{X_t > x \mid X_t > u(t), t\} = \left(1 + \xi \frac{x - u(t)}{\sigma(t)}\right)_+^{-1/\xi} \quad (\text{v8 §2.2, eq. (4), p. 3}).$$

That is the licence to model only the tail and to extrapolate beyond the data: the GP is not one convenient choice among many, it is the only continuous limit available. The threshold is our moving $u(t; 0.95)$, so the theorem is applied date by date.

Reading the two parameters. $\sigma(t)$ is the typical overshoot size at date t — a trend in it says floods are growing. ξ is the tail-weight: $\xi > 0$ heavy-tailed with no upper limit (Pareto-like); $\xi = 0$ exponential; $\xi < 0$ a *finite* upper end point at $u + \sigma/|\xi|$. Letting ξ move with t would claim the character of the tail is changing — a far stronger claim than growth — and ξ is the hardest parameter to estimate, so it is held constant per series (the Faulkner et al. decision).

Our numbers. Exceedances 2264 / 2236 / 2299. Shapes $\hat{\xi} = 0.038$ (NTHMB sf), 0.118 (BRIDG sf), -0.058 (BRIDG bf) — quote these when asked; with ~ 2250 exceedances the standard error is roughly $(1 + \xi)/\sqrt{n} \approx 0.025$ before clustering inflation, so all three are within a couple of standard errors of the exponential boundary $\xi = 0$: moderately light tails, no evidence of wild heavy-tailedness. Scale trends: likelihood-ratio statistics 108.2, 35.4, 89.3 on one degree of freedom — all far beyond any conventional threshold (χ_1^2 at the 0.001 level is 10.8). NTHMB’s mean scale by era: $29.0 \rightarrow 63.7$ — the typical overshoot more than doubles; BRIDG rises 8.5 to a mid-century 13.8 then eases to 12.0; baseflow $0.73 \rightarrow 2.1 \rightarrow 1.7$.

The two boxplots, and what each proves. Raw excesses by month show strong seasonality — freshet overshoots are physically larger — which is why σ must be seasonal. Excesses *divided by* the fitted $\sigma(t)$ should be month-free if the seasonal scale absorbed it: the boxes are level, which is the professor’s “excesses no longer seasonal” check passing in the body. The outliers are not level — winter months contain standardized excesses far larger than summer’s — which is the tail story below.

Question: “is the constant shape defensible?” Here is the honest answer, with its diagnostic. Refit the GP per season on the standardized excesses: DJF $\hat{\xi} = 0.271$ ($n = 478$), MAM 0.193, JJA -0.180 , SON 0.051. Winter is heavier-tailed than summer, and the same signal appears independently in the seasonal Exp(1) panels (the DJF excesses bow above the line) and in the standardized-excess outliers. Per-season standard errors are roughly 0.06 before clustering inflation, so DJF versus JJA is a real contrast, not noise — but freeing ξ by season is a modelling decision with real cost (short winter records, harder identifiability), and it should be made with the professor, not slipped in.

1.5 Station 5: the uniformity gate, in full

This is the gate of the whole analysis (v8 §3, p. 4), it is the check the professor named, and every follow-up question about it has a precise answer.

1.5.1 The theorem: why correct margins force a uniform histogram

The probability integral transform. If a continuous random variable X has distribution function F , then $U = F(X)$ is Uniform(0, 1). The proof is three lines: for $u \in (0, 1)$,

$$\Pr(U \leq u) = \Pr\{F(X) \leq u\} = \Pr\{X \leq F^{-1}(u)\} = F\{F^{-1}(u)\} = u,$$

and “ $\Pr(U \leq u) = u$ for every u ” is the definition of Uniform(0, 1). The intuition: $F(x)$ asks “what fraction of this distribution lies below x ?”, and a random draw lands below its own 30th percentile exactly 30% of the time, below its own 70th percentile exactly 70% of the time — at every level equally. Feeding a variable through its own distribution function erases the shape of the distribution and leaves pure rank.

The non-stationary version, which is our gate. Every day t owns a distribution $F(\cdot; t)$, and each day’s value is pushed through *its own day’s* fitted function: $U_t = F(x_t; t)$. If the fitted margins are correct, then U_t is Uniform(0, 1) for every single day — a June freshet day and a February low-flow day become identically distributed — and a mixture of uniforms is uniform, so the pooled sample of 47,847 values must look uniform. The contrapositive is the diagnostic: any deviation from flatness is an accusation that $F(\cdot; t)$ was wrong *for some identifiable set of days*, and 1.5.3–1.5.4 are the instruments that identify the set.

Three fine points that questions tend to probe.

1. *Continuity is load-bearing.* If the data carry ties, F has jumps and $F(X)$ has point masses — the histogram would spike at the tied values’ probabilities no matter how good the model. This is why the forensics of Station 1.1 run first: for streamflow there are no ties; baseflow’s 3% tie mass is the one place a small spike would be data, not model.
2. *Uniformity is marginal, not joint.* The theorem makes each day’s U_t uniform; it says nothing about independence, and indeed consecutive U_t are strongly dependent — a wet week is a week of high ranks. This does not move the target (flat at 1); it inflates the *noise* around the target, quantified next.
3. *The check is slightly in-sample.* The same days that built $\hat{F}$ are pushed through it, and a quantile fit calibrates its own rung shares by construction, so the pooled histogram flattens the fit a little at the rung levels. The sliced views, the era maps and the rung audit are what keep the check honest; a fully out-of-sample version (transform each year through a fit that excluded it) is a possible refinement, not currently needed.

1.5.2 The histogram: what a bar is, how flat is flat, and why 40 bins

What a bar is. With bins of width $w = 1/40 = 0.025$, the bar over a bin plots $\text{count}/(nw)$, so a perfectly uniform sample gives every bar height 1 *in expectation*. The red line at 1 is that expectation.

How much a bar may wobble. Under *independent* days the bin count is $\text{Binomial}(n, w)$ with $n = 47,847$: mean $nw \approx 1196$, standard deviation $\sqrt{nw(1-w)} \approx 34$, so the bar height has standard deviation $34/1196 \approx 0.029$ and a ± 2 -SD range of about ± 0.06 . But the days are not independent: flow persists, so the U_t move in runs of several days, and clustered counts have variance inflated by roughly the mean run length. With runs of 4–8 days the standard deviation multiplies by $\sqrt{4}$ – $\sqrt{8} \approx 2$ – 3 , giving honest fluctuation of roughly ± 0.15 per bar. *So “most bars between 0.85 and 1.15” is what a correct model looks like at this sample size*; demanding every bar touch 1 would be demanding less noise than 47,000 dependent days can deliver. The run-length arithmetic is an order-of-magnitude argument — which is why the report does not rely on it, but computes the noise band empirically (next).

Why 40 bins, and why not more. Bar noise scales like $1/\sqrt{nw}$: halve the width (80 bins) and the iid noise grows to ± 0.08 , with clustering to roughly ± 0.2 ; at 400 bins each bar holds ~ 120 days and the display is pure noise. Go the other way (10 bins) and the bars are quiet (± 0.03) but blind to any feature narrower than 0.1 in probability. Forty is a display-resolution choice, not a law: width 0.025 matches the spacing of the ladder’s own bottom rungs, so one bar roughly interrogates one rung interval. And the same theorem is checked at two other resolutions: the **QQ plot is the bin-free version** — it compares all 47,847 order statistics against their reference positions, no binning at all — and the **rung audit is the exact version** at the thirteen fitted probabilities: the share of days at or below $u(t; \tau_k)$ minus τ_k , at most 0.006 in absolute value everywhere in our run, with the only systematic entries at the bottom (-0.006 , -0.005 , -0.005 at $\tau = 0.01$: the 1% rung sits slightly low — the known difficulty of extreme quantile levels, and the cause of the short first bar).

1.5.3 The band figure: a noise interval on every bar

The question each blue segment answers: *how much would this bar move if history had dealt us a different set of years?*

```

STEP 1  compute the observed 40 bar heights once
STEP 2  repeat 200 times:
        draw 131 year labels with replacement from {1969, ..., 2099}
        keep the days of the drawn years           # whole years: the block
        recompute the 40 bar heights on this replicate
STEP 3  for each bar: the segment spans the 2.5% and 97.5% quantiles
        of its 200 replicate heights
STEP 4  flag (red dot) every bar whose whole segment sits off 1

```

Why whole years are the resampling unit. The noise to be measured comes from serial dependence — flood clusters, wet seasons — and a year is the natural block that keeps a full seasonal cycle and its clusters intact. Resampling single days would destroy exactly the dependence that creates the noise and produce a band that is too narrow (the iid ± 0.06), flagging noise as structure.

How to read the figure. Grey bars: observed heights. Blue segments: the year-block interval per bar. Red line: the target 1. Red dots: bars whose interval excludes 1 — deviations that no plausible re-draw of years repairs, i.e. structural; those are the bars the residual maps then attribute. Two honesty notes. First, $\hat{F}$ is not refit inside each replicate, so the band measures sampling variability of the days, not uncertainty of the fit — the right yardstick for “is this bar’s deviation year-to-year luck.” Second, the implemented draw keeps a drawn year once even when it is drawn twice, a subsampling flavour of the bootstrap that makes the bands slightly *wider* than the exact multiset version — the conservative direction for a diagnostic (it flags too few bars, never too many).

1.5.4 The residual maps: attributing every off bar

The construction. A contingency table: rows are slices (the 12 months, or the 6 eras), columns are the 40 bins, and the cell count O is “how many January days landed in bin $(0.025, 0.050]$.” The null hypothesis — margins correct — says each slice spreads *its own* days evenly over the bins, so the expected count is $E = (\text{row total})/40$. The colour of a cell is the Pearson standardized residual

$$R = \frac{O - E}{\sqrt{E}},$$

approximately standard normal under the null (row totals put E near 100 per month cell, near 200 per era cell): white ≈ 0 is perfect, red is too many days, blue is too few. $|R| \leq 2$ is ordinary noise; because serial clustering inflates cell noise and 480 cells bring multiplicity (the largest of 480 null residuals is around 3 by chance alone), the working rule is that $|R| \geq 4$ is structure — and the eye should look for *stripes*, not lone cells.

The reading rule — this is the point of the figure. A horizontal stripe along a *month* row means that month is systematically mis-transformed: residual seasonality, the one outcome the professor’s verdict forbids. A vertical stripe down a *bin* column means one rung is miscalibrated for all months alike: a ladder-tuning matter, not seasonality. A stripe along an *era* row means drift the slow basis missed. The map therefore does not just detect non-uniformity — it *classifies its cause*.

Our numbers. No month row shows a stripe — the seasonal complaint is answered. The largest cells: BRIDG, era [1969, 1991) at the top bin, $R = +15.2$ — too many first-era days counted as

extreme, the same first-era threshold miscalibration the local 5% table shows and the tensor design repairs; NTHMB, January at bin $(0.025, 0.050]$, $R = +7.1$, a bottom-rung neighbourhood effect consistent with the low 1% rung; NTHMB, era $[2013, 2035)$ at the bottom bin, $R = -7.5$, the same rung read by era.

What to say. “The PIT is uniform to within the sampling noise that 47,000 serially dependent days allow, in each season and each era separately; the report tests every bar against its own year-block noise band and attributes every genuine deviation — and what survives is not seasonal: it is the slightly-low 1% rung and the first-era threshold level, both identified and both with a named repair.”

1.6 Station 6: the dependence model and the model-free evidence

The model in words. On the common Laplace scale, write Y_C for the conditioning station and Y_X for the associated one. Given that Y_C is extreme ($Y_C = y > v$), the model says

$$Y_X = a(t)y + y^{b(t)}\{m(t) + s(t)Z\}, \quad Z \sim G$$

(v8 §5, eq. (8), p. 11). Read it as: when one river is extreme at rarity y , the other typically reaches the fraction $a(t)$ of that rarity, with scatter growing like $y^{b(t)}$, and the leftover randomness Z carries no formula — it is represented by the residuals the data themselves produced. The form is not arbitrary: Heffernan and Tawn showed that for essentially all standard dependence models, the correct normalisation of Y_X given $Y_C = y$ large is location ay and scale y^b — so the parametric part is the *shape theory dictates*, and everything data-driven lives in the four curves and the residual distribution.

Why a is the number that matters. At $a = 1, b = 0$ the model reads $Y_X \approx Y_C + Z$: asymptotic dependence — the link that never fades however extreme the event. $a = 0, b = 0$ is exact independence. Everything strictly between is asymptotic independence with strength a ; the Gaussian copula with correlation ρ sits at $a = \rho^2, b = \frac{1}{2}$. On the Laplace scale both tails are exponential, so a single form covers negative association too ($a \in [-1, 1]$; Keef et al. 2013). Because a is a function of t , the model can say the *class* of dependence is changing — which is the paper’s thesis, and Case 4 was selected because the screening said exactly that.

The model-free evidence, before any fitting. The windowed $\chi(0.95)$: in each six-year window, count the days on which *both* stations exceed their own pooled 95th percentile, divide by the days in the window, divide by 0.05. This estimates $\Pr\{\text{BRIDG extreme} \mid \text{NTHMB extreme}\}$ from ranks alone: independence gives $0.05/0.05 \rightarrow$ it hovers near $0.05/0.05 \cdot 0.05 = 0.05$ divided by 0.05, i.e. the value 0.05 shown as the dashed line after normalisation; perfect dependence gives 1. Ours climbs from 0.253 (mean of the first ten windows) to 0.403 (last ten). *The joint extremes are becoming more frequent, and no model was involved in saying so.* This is the yardstick every fitted dependence result must face.

1.7 Station 7: the conditional-simulation display, in full

This figure carries the most explanation per square centimetre, so it gets the same four-part treatment as the uniformity gate: what it tests, the generator with its distribution theory, how to read every visual feature, and the display decisions. If you can present this section, the functionals of Station 1.9 come free, because they are readings off the same generated cloud.

1.7.1 What the figure claims to test

The fitted dependence model is a machine for answering “given NTHMB is extreme, what does BRIDG do?” The pseudosample figure runs that machine *forward*: it manufactures synthetic days from nothing but the fitted parameters and the stored residuals, and overlays them on the real days. Two properties are then visible to the naked eye. *Agreement*: where data exist, the synthetic cloud should live where the data live — same centre, same spread, same tilt. *Extrapolation*: beyond the largest observed events the cloud keeps going, and that extension is precisely what the model would answer if asked about a 200-year event — shown, so its plausibility can be judged, rather than asserted inside an integral. This is Heffernan and Tawn’s own Figure 3 device, and it is the eye’s version of goodness-of-fit for a conditional model.

1.7.2 The generator, step by step, with the theory

Step 1: draw the conditioning value. We need Y_C from its distribution *given* $Y_C > v$. Two facts make this exact. First, after the marginal transform, Y_C is standard Laplace, whose upper tail is $\Pr(Y > y) = \frac{1}{2}e^{-y}$ for $y > 0$ — exactly exponential. The exponential is memoryless, so the excess above any threshold $v \geq 0$ is itself standard exponential: $\Pr(Y > v + w \mid Y > v) = e^{-w}$. Second, the inverse-CDF method: if U is Uniform(0, 1) then $\kappa + (1 - \kappa)U$ is Uniform(κ , 1), and pushing it through the Laplace quantile function gives

$$\Pr[q_{\text{Lap}}\{\kappa + (1 - \kappa)U\} \leq y] = \frac{F_{\text{Lap}}(y) - \kappa}{1 - \kappa},$$

which *is* the conditional CDF of Y_C given $Y_C > v$ (with $v = q_{\text{Lap}}(\kappa)$, $\kappa = 0.95$, $v = 2.303$). No approximation: the draw is from the fitted marginal law restricted to the exceedance region.

Step 2: draw the residual — and why the residual table exists. During fitting, for every historical day on which the conditioning station exceeded v , the model equation is solved backwards:

$$z = \frac{(y_X - a(t)y_C)/y_C^{b(t)} - m(t)}{s(t)},$$

i.e. subtract the fitted level, divide by the fitted growth factor, remove the residual location and scale. What remains is how far that day sat above or below its fitted expectation, in standard units, with the effects of rarity and date both removed. These numbers — one per historical extreme day, 2264 of them — *are* the estimate of G . No density is fitted to them, deliberately: a parametric G would smooth away exactly the tail irregularities that matter, while the empirical table keeps every skewness and heavy-tail feature the rivers actually produced. Generation samples one row with replacement. The licence for reusing a 1975 residual at a 2095 date is the model’s load-bearing assumption — after the four curves remove the covariate effect, the residual law no longer depends on t — and that assumption is testable and tested: the residual-by-era boxplots must show equal boxes, and the residual-versus- y_C scatter must show a flat running mean (both in the report).

Step 3: assemble at a chosen date. Evaluate the four curves at one day — reference year Y , day-of-year 187, the freshet peak — and compute $y_X = a y_C + y_C^b(m + sz)$. Every circle is one repetition of steps 1–3; the panels draw 600.

Step 4 (original-scale figure only): push back to flow units. A simulated y_X is converted to a probability $p = F_{\text{Lap}}(y_X)$ and then through the *inverse* of that day’s fitted marginal: below-threshold probabilities invert the ladder interpolation, $p > 0.95$ inverts the GP tail, $p < 0.01$ inverts the exponential lower piece. This is the same $F(\cdot; t)$ machinery as the transform, run backwards, so the marginal calendar (freshet high, winter low, later years shifted) is restored exactly.

1.7.3 The Laplace-scale panel, element by element

The panel titled “model pseudosample of (Y_C, Y_X) , year 2034.”

The axes. Both variables after the per-day marginal transform, in standard Laplace units — the units of *rarity for that day of that year*. Zero is a median day for its date; 2.30 is the top 5% for its date; +5 or +9 is far tail. The point of these axes: the calendar is gone — a February day and a July day, a 1970 day and a 2095 day, are directly comparable, because each was judged against its own day’s distribution.

The grey cloud (all 47,847 days). A diamond-shaped cloud centred at $(0,0)$ — the Laplace shape, exponential in both directions — tilted up-to-the-right: positive association surviving the transform, at all levels, not only extremes. Its symmetric, centred look is itself evidence that the margins were removed correctly (a wrong transform would displace or bend it). The upper reach ($\sim +11$) exceeds the lower (~ -5): extreme high flows go deeper into their tail than droughts do here.

The dashed vertical line at $v = 2.303$. The dependence threshold — the Laplace image of the 0.95 level, a *single flat number valid for every day*, possible only because the transform absorbed the non-stationarity. The model claims nothing left of this line, which is why no circles appear there: absence by construction, not failure.

The blue circles (600 generated pairs, year 2034). The year is the axis midpoint — the most stable region of the fitted curves; it enters only through a, b, m, s , since these axes carry no calendar. Their anatomy:

- *Dense at the line, thinning rightward:* the conditioning value is threshold plus an $\text{Exp}(1)$ draw — mass decays like e^{-w} beyond v — matching how the grey points themselves thin to the right.
- *Their centre rises only gently with x :* the gentle rise is $\hat{a} \approx 0.095$ under the current fit; the vertical width is s times the residual table’s spread, widening slowly because $\hat{b} \approx 0.43$.
- *Circles at low y , even below zero:* not an error — the finding. Asymptotic independence means BRIDG can be an unremarkable day while NTHMB is extreme, and the grey data right of the line show the same low points. A model forcing every circle high would be claiming asymptotic dependence, which this pair does not have.
- *The occasional very high circle (near $(5,9)$):* a heavy residual drawn from the table — the empirical G kept the rivers’ real skewed, heavy residuals, which a fitted Gaussian would have erased.
- *The agreement test, by eye:* compare circles with grey points in the same x -range — same centre, same spread, same skew. Circles reaching $x \approx 8-9$, where the grey points run out, are the extrapolation, on display.

The honest caveat belonging to this panel. These circles were generated under the dependence fit that Section 1.18 shows never left its starting value ($\hat{a}$ flat at 0.095). The cloud still overlays the data credibly because m, s and the empirical residuals absorb much of what a mis-estimated a leaves behind — which is exactly why the eye test alone cannot settle a -versus- m , and why the optimiser repair plus the bootstrap, not this picture, will carry the trend claim.

1.7.4 The original-scale figure, element by element

The figure titled “model pseudosamples on the original scale, day-of-year 187, years 1975 and 2095,” with both axes logarithmic.

Why log axes. The flows span three orders of magnitude (BRIDG 0.5 to ~ 300 , NTHMB 10 to ~ 1500); linear axes would crush the cloud into a corner. On log axes equal *ratios* are equal distances — so a cloud that looks compact on the right actually spans a wide absolute range.

The grey cloud. Three features worth naming. It rises diagonally: association at all levels. It is made of fine *parallel threads*, clearest lower-left: recession curves — consecutive days of one storm or melt event trace a smooth trajectory as both rivers drain, one thread per event, made visible by the log scale. And it has *broad branches* rather than one ellipse: rising-limb, falling-limb and different seasons occupy slightly different tracks (hysteresis) — a reminder that the body of the joint distribution is complicated, which is why the model only claims the conditional tail.

The black triangles (one per year). For each of the 131 years, the day NTHMB hit its annual maximum, plotted at that day’s (NTHMB, BRIDG) pair; they pile up at NTHMB ≈ 500 –1100, BRIDG ≈ 80 –250. They exist for one reason: the meeting’s complaint was that simulated points came out smaller than real extremes, so the real extreme days are put on the plot and the clouds must land on and extend their territory — checkable by eye. (The stray low triangle near (350, 40): a weak flood year, real, not an error.)

The blue circles (year 1975, day 187). Each is one run of the generator with both coordinates pushed back through *that day’s* inverse marginals. Two visible properties: the column is *narrow in x* — the conditioning draw concentrates just above the threshold (exponential decay of mass) and 1975’s day-187 margin maps that concentration into a tight band starting at the day’s 95th percentile (~ 650); and its *vertical spread* (~ 80 –250) matches the triangles’ spread — the calibration check passing.

The red circles (year 2095, same day). Everything that differs from blue is the fitted change across the century: the cloud shifts *right* (NTHMB’s margin intensifies — threshold slope +61.8 per unit axis, GP scale $29 \rightarrow 64$), stretches into territory with no grey dots — the extrapolation, displayed — and rises slightly. Honest note to volunteer if pressed: with the current stuck fit $a(t)$ is flat, so nearly all visible blue-to-red movement is the *margins* changing; the dependence component joins after the optimiser repair.

One-sentence summaries. Laplace panel: “on the common rarity scale, the model’s synthetic extreme days — parameters plus recycled residuals only — are statistically indistinguishable from the real days where data exist, and beyond the data they show exactly the extrapolation the model would use for rarer events.” Original scale: “the same synthetic days placed on the real flow scale at the freshet peak of an early and a late year: they sit where the real annual-maximum days sit, and the red cloud is what the fitted century of change does to a co-flood.”

1.7.5 The display decisions, each with its reason

Why the conditioning day is the freshet peak (day 187). Conditioning on a randomly picked date lands mostly in low-flow seasons; the back-transform then returns modest flows, and the simulated points come out smaller than the real annual extremes — precisely the complaint raised in the meeting. Real annual maxima occur on particular days of the season, so a display meant to be compared with them must condition on the day of year. The original-scale figure

overlays the observed annual-maximum days as black triangles so the comparison is checkable by eye: the clouds now land on and extend the triangles’ territory.

Why no year colours on the standard-scale panel. The marginal transform absorbs the year: after each day passes through its own day’s distribution, all days follow the same Laplace law, so 1973 and 2094 are *by construction* indistinguishable on those axes, and colouring them would draw a distinction the axes cannot carry. (The joint drift, if any, lives in $a(t), b(t)$ — and is quantified by the fitted curves and the functionals, not by colours here.)

Why the year contrast is legitimate on the original scale. There the margins genuinely differ by year — the same Laplace pair maps to different flows in 1975 and 2095 — so blue-versus-red carries real information: the whole fitted change, margins and dependence together. Which scale the paper should show is the open display question for the professor; the eventual headline is RCP 4.5 against RCP 8.5 on this same figure, once that export is in the pipeline.

Why years 1975, 2034 and 2095 — deliberately not extreme years. A display date has two coordinates chosen by different logic. The *day-of-year* coordinate (187) is chosen to be extreme, on purpose and by computation, not by hand: it is the day at which the fitted threshold’s climatological average peaks, so the simulation is comparable with real annual maxima. The *year* coordinate is structural, not extreme: `start_year + 6, round(mean(range(years)))` and `max(years) - 4` give 1975, 2034 and 2095 — early, middle and late positions fixed by arithmetic before any result is seen. Why those positions: early-versus-late is the figure’s job (same conditioning event, same day of year, only the year differs, so the blue-to-red difference is purely the fitted change); a few years *inside* the axis ends because spline fits are least stable at boundaries; the middle year for the single-cloud panel because it is the most stable region and, on the standard scale, any year works almost by construction. The distinction matters: picking a year *because it contained a famous flood* would be selection bias — choosing the display after seeing outcomes; axis positions declared in advance are illustration (v8 §6: “choosing a date to display is illustration, not selection bias, provided the questions the figure answers are written down first”). And the conditioning *value* is never chosen at all — it is integrated over: Y_C is drawn from the whole top-5% region, so the display conditions on “extreme,” not on one historical event.

Monte Carlo error, in case it is asked. With 600 circles the cloud’s visual boundary wobbles; with the $n_{mc} = 10,000$ draws used for the functionals, a probability near 0.35 carries a Monte Carlo standard error of $\sqrt{0.35 \times 0.65/10,000} \approx 0.005$ — negligible against everything else.

1.8 Station 8: return levels

The definition, and why it needs a reference year. Under a trend there is no single “100-year flood.” Fix a station, a period T and a reference year Y : freeze the fitted curves as they stand in year Y , let M_Y be the annual maximum of a year with those parameters, and define $x_T(Y)$ by $\Pr\{M_Y > x_T(Y)\} = 1/T$ (v8 §6, eq. (10), p. 14). Every return-level statement is a probability statement about a year with a stated climate — the waiting-time reading (“we will wait about T years”) is false under a trend and is never used.

The construction, in three moves. (i) For x above the threshold on day t , the daily exceedance probability is the chance of being an exceedance day at all times the GP tail beyond x : $p(x, t) = 0.05 \times \{1 + \xi(x - u(t))/\sigma(t)\}_+^{-1/\xi}$ (v8 eq. (11)). (ii) Chain the days of year Y : $\Pr(M_Y \leq x) = \prod_t \{1 - p(x, t)\}$ — exact if days were independent. (iii) They are not: extremes arrive in clusters, so the year holds fewer independent opportunities than days, and the classical extremal index result

says the dependent maximum behaves like the independent one raised to the power $\theta \in (0, 1]$: $\Pr(M_Y \leq x) \approx [\prod_t \{1 - p(x, t)\}]^\theta$ (v8 eq. (13), p. 15), where $1/\theta$ is the mean cluster length. Ignoring θ overstates the annual exceedance probability and *inflates* the return levels — wrong even though it errs on the cautious side.

Our numbers. $\hat{\theta} = 0.244$ at NTHMB (mean cluster 4.1 days), 0.127 at BRIDG (7.9 days), with our implementation and `extRemes` agreeing to three decimals. The two computational routes — root-finding on the formula versus simulating thousands of synthetic years and reading the empirical quantile — agree to 0.26%, 0.32%, 0.98% at $T = 10, 50, 100$ when clustering is switched off, which validates the chaining machinery before the correction is trusted.

1.9 Station 9: the three conditional functionals

The professor’s three summaries — the conditional mean $\mathbb{E}(Y_t | X_t \geq x)$, the conditional quantiles, and the band probability $\Pr(a \leq Y_t \leq b | X_t \geq x)$ — are all read off one generated cloud per reference year, produced by exactly the generator of Station 1.7 with $n_{mc} = 10,000$: the mean is computed *after* inverting to flow units (the mean does not commute with the nonlinear transform, so it must be taken on the flow scale); the quantiles are empirical quantiles of the generated flows; P_{co} — BRIDG above its own day’s threshold while NTHMB is extreme — needs no inversion at all, because on the Laplace scale the day’s threshold is the same flat number $y^* = 2.303$ for every day, so it is just the fraction of the cloud above y^* ; likewise P_{band} with the band bounds at the day’s marginal 0.95 and 0.995 quantiles, which on the Laplace scale are the fixed numbers 2.303 and 4.605. All bounds and levels are declared parameters, because they were the professor’s free choices.

Status. The machinery is verified — the model-generated conditional mean matches the plain empirical average over observed exceedance days to 0.5% (early era) and 8.4% (late era), the HT04 Table-5 discipline — but the values currently printed ($P_{co} \approx 0.35$, flat in t) are computed from the stuck dependence fit of Section 1.18 and must be re-generated after the repair. Present the machinery, not the numbers.

1.10 Station 10: uncertainty

Whole years are resampled with replacement, keeping each year’s days together and in order — the block that preserves within-year clustering — and threshold, GP tail, extremal index and dependence model are *refitted* inside every replicate; the spread of the recomputed quantities is the uncertainty band. This is different in kind from Station 5’s band (years re-drawn, nothing refitted: a sampling-noise yardstick) and from Route B (generation from a fitted model: Monte Carlo, not uncertainty). The knit runs $B = 25$ replicates for speed; the paper-scale run is $B = 2000$ with every data-driven selection (smoothing weights, model sizes, the lag) re-selected inside each replicate — selection outside the loop makes bands too narrow, because it treats a chosen quantity as if it had been known in advance.

Part II — Inside the code

Every algorithm: what it computes, the pseudocode, why that is the right computation, and what verifies it. Where a published package does the work, what it does internally is explained — “we use qgam” is not an explanation.

1.11 The one formula behind every parameter curve

Every parameter — threshold, scale, the four dependence curves — is one formula with different ingredients:

$$\eta(t) = \sum_{k=1}^p \gamma_k B_k(t),$$

where the B_k are fixed, known basis functions chosen before fitting and the γ_k are the only estimated quantities (v8 §4.1, eq. (7), p. 5). One column: a constant (the stationary model). Two columns $[1, t]$: a straight line. Cubic B-spline bumps at knots along the axis: the flexible curve. The bases are nested — constant $\subset$ line $\subset$ spline — which is what makes model-size comparison meaningful. A cubic B-spline bump is nonzero over only four knot intervals, rises to one peak, and returns to zero; at any date exactly four bumps are nonzero and they sum to one, so the curve at a date is a local weighted average of four coefficients, each estimated from the thousands of days under its own bump. Neighbours replace repeats; smoothness is the assumption that makes that legitimate. The seasonal component adds cyclic basis functions in day-of-year (period one year, derivatives matching across the year boundary); the tensor version multiplies the two bases into a surface.

1.12 The quantile ladder in code: qgam, line by line

The target is unchanged from Station 1.2: for each level τ in $\{0.01, 0.025, 0.05, 0.1, \dots, 0.9, 0.95\}$, the curve minimising pinball loss plus roughness — and minimising pinball loss *is* quantile estimation, by the derivative identity proved there. What follows is every qgam-touching line, in the code-understanding format.

Entry 1: the model formula.

```
fm <- list(y ~ s(doy, bs = "cc", k = 12) + s(tt, bs = "cr", k = 25),
           ~ s(doy, bs = "cc", k = 8))
```

`s()` declares a smooth term in mgcv's language. `bs = "cc"` is the *cyclic* cubic spline: the basis is built so the curve and its derivatives match across the year boundary — December 31 joins January 1 with no seam; one seasonal shape, learned from all 131 years at once. `bs = "cr"` is the ordinary cubic spline for the trend; $k = 25$ over 131 years is a knot roughly every 5.5 years, matching the project's spacing decision. **What k is and is not:** k is the basis dimension — a *ceiling* on flexibility, not the flexibility itself; the actual wiggleness is chosen by the smoothing penalty (Entry 3), so a generous k with automatic smoothing is the standard idiom. **The second formula is not a second quantile model:** it tells qgam how the conditional *spread* of y varies, because the smoothed loss has a per-observation bandwidth that must scale with local spread (Entry 3, step 1). Flow variance is massively seasonal — freshet days scatter over hundreds of units, winter days over a few — so the variance model is cyclic in day-of-year. Omitting it is qgam's classic failure mode: a constant bandwidth gives the fitted 0.95 curve season-dependent bias (their own motorcycle-data figure demonstrates it).

Entry 2: the pass-through knots.

```
argGam = list(knots = list(doy = c(0.5, 366.5)))
```

Everything in `argGam` goes to the internal `mgcv::gam` calls. This pins the cyclic boundary knots at half-days: the period is exactly 366 days with the join *between* Dec 31 and Jan 1. Without it, mgcv sets the cycle boundary at the observed data range (1 to 366), which would identify day 1 *with* day 366 — gluing New Year onto New Year's Eve as the same covariate value instead of neighbours.

Entry 3: the fit itself.

```
fitM <- mqgam(fm, data = d, qu = taus, argGam = kn)
```

`mqgam` fits the model at all 13 levels, storing the shared components once (why it beats 13 `qgam` calls). Internally, in order:

1. *Preliminary location-scale fit.* A Gaussian `gaulss` GAM of mean and variance (the variance from our second formula), plus a flexible density (`sinh-arcsinh`) for its standardized residuals. Purpose: the pinball loss has a kink, which blocks smooth optimisation; `qgam` replaces it with the smoothed *ELF loss*, and the amount of smoothing is *computed, not tuned* — a formula minimising the asymptotic mean squared error of the coefficients, evaluated with this density estimate, giving a *per-observation* bandwidth: wide in the freshet, narrow in winter. Their theoretical result: the optimally smoothed loss is statistically *better* than the raw pinball, not merely convenient.
2. *Learning-rate calibration, per level* — the expensive step and the printed dots (“**Estimating learning rate... qu = 0.5done**”). The rate $1/\sigma_0$ balances loss against the smoothing prior; `qgam` selects it by Brent search where *every dot is a complete penalized fit* at a trial σ_0 , scored so that the intervals for the quantile curve achieve roughly their advertised coverage. This is why 13 rungs $\times$ 5 series was slow: calibration, not fitting.
3. *Smoothing selection.* Given σ_0 and the bandwidth, the roughness weights λ (one per `s()` term) are chosen by Laplace-approximate marginal likelihood — the step that replaced our cross-validation loop at this stage (the head-to-head is Section 1.13).
4. *Final fit* at the chosen σ_0, λ by penalized iteratively re-weighted least squares — `mgcv`’s standard engine.

Recorded costs, all handled by our code: no non-crossing constraint (Entry 5); calibration theory assumes independent data, so with 4–8-day clusters `qgam`’s intervals are a working approximation and *all* our uncertainty comes from the year-block bootstrap; and extreme- τ bias — our rung audit shows exactly this cost, the $\tau = 0.01$ rung sitting $\sim 0.5\%$ low.

Entry 4: extracting the ladder.

```
U <- sapply(taus, function(q) as.numeric(qdo(fitM, q, predict, newdata = d)))
```

`mqgam` stores its 13 fits stripped of shared components to save memory, so they cannot be used directly; `qdo` re-attaches the shared pieces to the level- q fit and applies any standard gam function — here `predict` on all 47,847 days. `sapply` stacks the 13 curves into the $N \times 13$ matrix U . Downstream, *only U is used* — which is what makes the `qgam` route and the fallback interchangeable.

Entry 5: the non-crossing audit and repair.

```
ncross <- sum(apply(U, 1, function(r) any(diff(r) <= 0)))
if (ncross > 0) { Us <- t(apply(U, 1, sort))
  msize <- max(abs(Us - U)) / sd(x); U <- Us }
```

Per day, check the 13 rung values are strictly increasing; count the days with any violation; repair by sorting each day’s rungs into order — monotone rearrangement, which provably never increases estimation error (Chernozhukov, Fernández-Val, Galichon 2010). `msize` records the largest change the sort made, in sd units — the number separating “hairline overlaps” from “real fitting problem.” Our run: 1147 / 1520 / 1481 crossing days, largest repair 0.013 / 0.006 / 0.009 sd — cosmetic.

The returned `lam_u = NA` marks that `qgam` kept its smoothing internal; downstream code checks for the NA.

Entry 6: the shell — fallback and cache. The `tryCatch` around the body degrades to the project `quantreg` fitter on any error, announced with `cat("NOTE: ...")` — `cat`, not `message()`, because chunk options suppress messages and a silent fallback once hid a wrong route. The cache wrapper saves the result to `cache_ladder_<tag>_<route>_<N>_<sum(x)>.rds`: the name carries the route and a data fingerprint, so changing the data, axis or route changes the filename and forces a refit automatically; `refit_ladders: true` forces one manually; the knit prints which cache file it used — provenance on the record.

Entry 7: the fast-calibration variant.

```
cal <- tuneLearnFast(fm, data = d, qu = c(0.05, 0.50, 0.95), argGam = kn)
lsig_of <- approxfun(c(0.05, 0.50, 0.95), cal$lsig, rule = 2)
U <- sapply(taus, function(q) as.numeric(predict(
  qgam(fm, data = d, qu = q, lsig = lsig_of(q), argGam = kn),
  newdata = d)))
```

Entry 3’s step 2 runs per rung and dominates the cost. `tuneLearnFast` runs *only* that calibration, at three anchor levels; `approxfun` interpolates the selected $\log \sigma_0$ linearly in τ (the learning rate varies slowly and smoothly in τ , which is what makes interpolation safe); passing `lsig` to `qgam` skips calibration entirely, so each rung is one plain fit. Net: 3 calibrations + 13 fits instead of 13 calibrations — a 3–4× cut, same model.

Entry 8: the tensor experiment.

```
fte <- qgam(list(y ~ te(doy, tt, bs = c("cc","cr"), k = c(10, 15)),
  ~ s(doy, bs = "cc", k = 8)),
  data = d, qu = 0.95, argGam = kn)
```

`te()` replaces “seasonal shape + trend” with a smooth *surface* in (day-of-year, year): the marginal bases (10 cyclic × 15 trend) are multiplied into 150 surface functions with *separate* penalties in each direction — two smoothing parameters, both by Entry 3’s machinery — cyclic in the `doy` direction only. The additive model is a special case of the surface, so nothing can be lost, and the smoothing shrinks back toward additivity unless the data pay for the interaction. Fitted at the single working level $\tau = 0.95$ (one calibration), cached under its own filename. This is the fit whose season × era table turned the first-era misses (0.149, 0.123) into 0.039 — the evidence for adopting it.

1.13 qgam versus blocked cross-validation: one problem, two philosophies

Both routes estimate the same curve under the same loss; they differ in the criterion for one nuisance number, the roughness weight. The big picture in one sentence: *cross-validation treats smoothing selection as a prediction experiment — physically hold data out, measure pinball loss on it, keep the λ that predicts best; qgam treats it as Bayesian inference — translate the penalized problem into likelihood-plus-prior, then keep the λ under which the observed data are most probable*. Everything `qgam` builds (the smoothed loss, the learning rate) exists to make that translation legitimate, because a raw loss is not a likelihood and Bayesian machinery may not touch it directly. Step by step:

Step 0, the target: identical. The τ -quantile curve, defined by the pinball identity. Nothing about the route changes what is estimated.

Step 1, the objective. CV route: raw pinball + $\lambda \times$ squared differences of neighbouring coefficients. qgam: the same with the kink rounded off (ELF), the rounding set by the MSE-minimising formula with a per-observation bandwidth — not a new knob, and provably an upgrade, not a compromise.

Step 2, the inner fit at fixed λ . CV route: a linear program (`rq.fit.fnb` on the augmented design, penalty rows appended as pseudo-data). qgam: with a smooth loss, $\exp(-\text{loss})$ behaves like a density, so the fit is penalized likelihood via iteratively re-weighted least squares. Same role, different solver, enabled by step 1.

Step 3, choosing λ — the head-to-head. CV route: a grid of λ times five year-block folds; fit on four fifths, score held-out pinball, average, argmin. Direct and model-free, but expensive, one λ only (a two-dimensional grid times folds is unaffordable — which is why our CV shares a single λ across curves), and it stops at the grid edge (we hit that edge twice, at 10^4 and 10^6). qgam: the penalty is reinterpreted as a Gaussian *prior* — “wiggly curves are improbable a priori,” with λ its precision; integrate the coefficients out and what remains is the *marginal likelihood of λ itself*: how probable the observed data are under each candidate smoothness, averaged over every curve that smoothness allows. Laplace-approximate the integral (LAML), maximise by Newton. No grid, no folds, no edges, and *several* λ ’s at once (seasonal, trend, the tensor’s two). The bridge to what you know: maximising marginal likelihood is asymptotically close to optimising leave-out predictive performance — the known REML–CV connection — it reaches the same place by probability calculus instead of data splitting.

Step 4, the step CV does not have: the learning rate. A true likelihood fixes the relative weight of data and prior by itself; a loss-based pseudo-likelihood has no natural scale — $\exp(-\text{loss}/\sigma_0)$ is equally valid for any σ_0 , and σ_0 controls wiggleness and interval width. This is the price of the loss-to-likelihood translation (the Bissiri–Holmes–Walker belief-updating framework is its licence). qgam pins σ_0 by coverage calibration (the Brent search, the printed dots). Our CV route never needed this step because it never claimed intervals — uncertainty was always the bootstrap’s job; accordingly, the calibrated intervals are the one qgam output we deliberately do not use.

Step 5, serial dependence. Our blocked folds exist *because* of dependence: random folds leak a held-out day’s neighbours into training and under-smooth. LAML has no folds, so no leakage of that form — but it is not immune either: it counts every day as an independent contribution, overstating the effective sample size, with the same qualitative risk and definitely too-narrow intervals. *Neither criterion handles our dependence honestly by itself*; on both routes the final authority is out-of-model — the season $\times$ era tables — and the year-block bootstrap for uncertainty.

Step 6, what stays ours on either route. The rung grid, the basis design, the non-crossing audit and repair, the assembly into $F(x; t)$, and the verification ladder — including the comparison of the two routes themselves.

Step	Our CV route	qgam’s replacement
Loss	raw pinball	smoothed pinball (ELF); bandwidth by MSE formula + variance model
Inner fit	linear program (rq, augmented design)	penalized IRLS (loss now differentiable)
Penalty’s meaning	roughness term added to the loss	Gaussian <i>prior</i> on coefficients
λ selection	grid $\times$ year-block folds, held-out pinball	marginal likelihood (LAML), Newton, several λ ’s at once
Data-vs-penalty weight	implicit, absorbed in λ	explicit learning rate σ_0 , set by coverage calibration
Dependence	blocked folds	nothing built in — bootstrap and out-of-model tables cover it
Uncertainty	block bootstrap (external)	calibrated intervals (built in; deliberately unused here)

The honest closing symmetry, if asked “which is right”: neither, by authority. Where the two routes disagree — top-rung mean gap 0.166 sd — the referee is not preference but the out-of-model calibration tables, where the qgam route with the tensor design currently wins.

1.14 Blocked cross-validation — where it still runs, and why blocked

qgam and evgam select their own smoothing; cross-validation remains only where no package covers our criterion (our GP fallback fitter; the dependence stage). It selects one number, λ :

```

STEP 1  assign whole YEARS to K = 5 folds, round robin (1969->1, 1970->2, ...)
STEP 2  for each candidate lambda in a grid:
          for each fold k: fit on the other folds, score the pinball or
          negative log-likelihood on fold k alone; average the K scores
STEP 3  keep the lambda with the smallest average held-out score

```

Whole years, because flood days come in 4–8-day runs: with random-day folds the neighbours of a held-out day sit in the training set carrying nearly the same information, the held-out score is optimistic for wiggly fits, and CV systematically under-smooths. Blocking puts an entire cluster on one side of every split. Limitation, stated up front: CV answers “what predicts best” and can never certify a trend — the penalty’s null space (constants under a first-difference penalty) is invisible to it — so significance belongs to move one, never to λ .

1.15 Model size: the two moves

Move one, constant versus linear — the only step that can say “a trend exists.” The test matches the stage’s criterion. GP tail: a genuine likelihood, so the likelihood-ratio statistic $\Lambda = 2(\hat{\ell}_{\text{lin}} - \hat{\ell}_{\text{const}})$ is compared to χ_1^2 (Wilks: under the null, twice the log-likelihood gain from one extra free parameter is χ_1^2). Threshold: no likelihood exists, so the slope is judged against its block-bootstrap standard error. Dependence: the likelihood is a working device, so its Λ is only a guide and the honest test is the bootstrap interval for the slope. **Move two, linear versus spline**: no separate test — the penalised spline contains the line, so if the fitted spline collapses to a straight shape, the line is reported and the spline kept as robustness.

1.16 The GP tail fit

```

INPUT  excesses (x - u(t)) on the ~2250 days above the threshold
MODEL  log sigma(t) = cyclic seasonal + trend spline    (log => sigma > 0)

```

xi = one constant per series
FIT maximise sum of GP log-densities + roughness penalty on the trend

Production: `evgam, family = "gpd"`, REML smoothing. Our own penalised-likelihood fitter runs alongside for the constant-vs-linear LRT and as the consistency check: relative gaps 0.185 / 0.363 / 1.010 — the large baseflow value is two smoothing philosophies disagreeing where σ is tiny (a relative measure inflates), reported not hidden.

1.17 From two pieces to one distribution; the Laplace map; the inverse

$F(x; t)$ is assembled per day (v8 §3, eq. (5), p. 3): linear interpolation between rung values in the body; the GP tail above the top rung (continuous by construction, since $F = 0.95$ at the threshold on every day); an exponential lower piece below the bottom rung, slope-matched so the lower tail decays instead of clipping (the old hard clamp at 10^{-5} collapsed the lowest days onto one point at -2.44 and drew a flat-run-and-jump QQ artifact; the instrumented rebuild prints 0 days at the numerical floor). Then the Laplace map $y = \log(2F)$ for $F < \frac{1}{2}$, $-\log\{2(1 - F)\}$ otherwise (v8 eq. (6), p. 4). Why Laplace: both tails exactly exponential and symmetric, so one parametric dependence form covers positive and negative association ($a \in [-1, 1]$), where the Gumbel scale of the 2004 paper covers positive only. The inverse $q_{\text{inv}}(p; t)$ runs the same assembly backwards (ladder inverse in the body, GP quantile above, exponential inverse below) and is what returns simulated values to flow units.

1.18 The dependence fitter, the Keef constraint, and the defect

The objective. Writing $\mu_i = a(t_i)y_{C,i} + m(t_i)y_{C,i}^{b(t_i)}$ and $\sigma_i = s(t_i)y_{C,i}^{b(t_i)}$, minimise over the exceedance days

$$\sum_i \left[\log \sigma_i + \frac{1}{2} \{ (y_{X,i} - \mu_i) / \sigma_i \}^2 \right] + \text{roughness penalties},$$

the negative log of a working Gaussian likelihood. The Gaussian is a scoring device, not a belief — Keef et al. tested Normal against Laplace working residuals and kept the Normal — and the residual is stored as a table, never believed normal. Links: $a = 2 \text{plogis}(\eta_a) - 1 \in (-1, 1)$; $b < 1$ enforced; $s = e^{\eta_s} > 0$.

The Keef constraint (v8 §5, eq. (9), p. 13). An unconstrained fit can extrapolate conditional quantiles *above what perfect positive dependence would give* — a self-contradiction. The ordering constraint forbids it, checked at a level v_K above the largest observed conditioning value, because extrapolation is what needs policing.

The defect, precisely. The constraint was implemented as a hard wall: objective = 10^{10} outside the feasible set. BFGS estimates gradients by finite differences; when trial steps land on the wall the gradient is garbage and the search stops at its starting point. Evidence: all three model sizes returned the identical objective 2698.6; the trend statistic was exactly 0.00 with $\hat{a}$ flat at 0.095; switching the wall off dropped the objective to 2022.2 (a 676-unit gap — not a constraint biting mildly) and produced $\hat{a}(t) : 0.396 \rightarrow 0.490$, rising. So the reported constrained fit is the starting value, not an optimum. One corollary: *every* constrained linear/spline number in this knit is suspect — including the freshet-only linear endpoints ($0.443 \rightarrow -0.245$), fitted through the same wall; the freshet number that is reliable is the constant $\hat{a} = 0.227$.

What the answer probably looks like. The unconstrained trend ($0.40 \rightarrow 0.49$) and the model-free windowed χ ($0.253 \rightarrow 0.403$) agree in level and direction. And the constant fit — which was

seeded sensibly — agrees with `texmex`’s constrained reference (a : 0.095 vs 0.081; b : 0.425 vs 0.402), so the constraint machinery is sound in principle; only the wall is unusable by a gradient method.

The repair.

```
violation(a,b) = size of the ordering breach (0 when feasible)
objective      = working criterion + penalties + kappa * violation^2
fit            = Nelder-Mead first (derivative-free), BFGS to polish,
                from several starts incl. the unconstrained solution
report        = the violation at the solution (zero = feasible, printed)
```

The second finding, independent of the defect. Constant fits per season of the conditioning day: JJA $\hat{a} = 0.172$, MAM 0.141, SON -0.011 , DJF -0.055 . The association lives in the freshet; winter “extremes” (extreme *for the season*) carry none. Because the seasonal threshold spreads conditioning days across all seasons (537–599 per season), a pooled $\hat{a}$ mixes these regimes and is diluted toward zero — the freshet-only constant is 0.227 against the pooled 0.095. Whether the science question is season-relative extremes or flood-season extremes is a decision for the professor; the numbers show it changes the answer.

1.19 The verification ladder, in one place

The answer to “how do I know the code is right” — six independent checks, all printed in the report:

1. `qgam` ladder vs our penalised `quantreg` ladder: 0.166 sd.
2. `evgam` tail vs our penalised likelihood: 0.185 / 0.363 / 1.010.
3. Our all-constant dependence fit vs `texmex` on its own defaults (Laplace margins, Keef constraints), both given the identical transformed pair so the margins cancel: a 0.095 vs 0.081, b 0.425 vs 0.402.
4. With b frozen and curves linear, our fitter vs `mgcv::gam(gaulss)` on literally the same model: conditional means agree to 3.4×10^{-5} of the data range.
5. Route A root-finding vs Route B simulation: $\leq 1\%$.
6. The known-answer case: the Case 1 anchor pair must return $\hat{a}$ near 1; it returns 1.000.

Plus `extRemes` $\equiv$ our Ferro–Segers to three decimals, and the HT04 Table-5 check: model-generated vs empirical conditional mean, 0.5% and 8.4% by era.

Part III — Question drill

“Why quantile regression for the threshold?” A fixed level is a different rarity in different seasons and decades, so its exceedances are not comparable events; fixing the probability and moving the level makes “extreme” mean rarity 0.05 on every date — and minimising the pinball loss *is* quantile estimation, by the derivative identity $\frac{d}{du} \mathbb{E} \rho_\tau = F(u) - \tau$.

“How is the smoothing chosen?” Automatically: marginal likelihood with a calibrated learning rate at the threshold (`qgam`), REML at the tail (`evgam`), blocked cross-validation where no package covers the criterion. The hand-set quantities are declared design choices: the rung grid, knot spacing, basis sizes.

“Why whole-year folds and blocks?” Days cluster in 4–8-day runs; random folds leak a held-out day’s neighbours into training, bias the score, and under-smooth. Years are the natural block containing whole clusters and whole seasonal cycles.

“Why should $F(x_t; t)$ be uniform?” The probability integral transform: a continuous variable through its own CDF lands below its u -quantile with probability exactly u — the definition of uniformity. Each day uses its own day’s fitted CDF, and a mixture of uniforms is uniform (Section 1.5.1).

“Why 40 bars / why are bars not all at 1?” Resolution choice; bar noise $\propto 1/\sqrt{n \times \text{width}}$: ± 0.06 iid, $\approx \pm 0.15$ with clustering. Bars are tested against year-block resampling intervals, not against the eye (Sections 1.5.2–1.5.3).

“What is the heatmap?” Pearson residuals $(O - E)/\sqrt{E}$ of a slice-by-bin table; month stripe = seasonality, column stripe = a rung, era stripe = drift. Ours shows no month stripe (Section 1.5.4).

“What exactly is a ?” The fraction of the conditioning station’s rarity that the other station typically attains on a common standardised scale: 1 asymptotic dependence, 0 independence, Gaussian copula at ρ^2 ; time-varying a is the paper’s question (Section 1.6).

“Where do the simulated points come from?” An exact draw from the fitted marginal restricted to the exceedance region (Laplace tail is memoryless), a residual recycled from the historical table, assembled through the fitted curves at a chosen date, optionally inverted to flow units — Section 1.7, four steps, each with its theory.

“Why do the circles only appear right of the dashed line?” The model is defined only for conditioning values above v ; the left region is not modelled, so no circles belong there.

“What would a bad fit look like in the pseudosample figure?” Circles systematically above or below the data cloud (wrong a or m), circles fanning wider or narrower than the data (wrong b or s), or a symmetric cloud against visibly skewed data (a parametric G would do this; the empirical table is why ours does not).

“Why condition on day-of-year 187?” Because annual extremes occur on freshet days; conditioning on a random date compares simulated low-season days against observed annual maxima and loses — the complaint from the meeting, fixed by design and checkable via the overlaid triangles.

“Why years 1975, 2034 and 2095?” The day of year is chosen to be extreme — the climatological freshet peak, computed from the fitted threshold — so the simulation is comparable with real annual maxima; the years are early, middle and late axis positions fixed by arithmetic in advance. Representative dates, not selected events: picking a year for its famous flood would be selection bias, and changing these is a one-line parameter.

“CV or qgam — which is right?” Both estimate the same curve under the same loss; they differ only in the criterion for one nuisance number (Section 1.13). Where they disagree — 0.166 sd at the top rung — the referee is the out-of-model calibration table, where the tensor-qgam route wins.

“Is ξ constant defensible?” A decision, with the diagnostic on the table: per-season $\hat{\xi} = \text{DJF } 0.271, \text{ MAM } 0.193, \text{ JJA } -0.180, \text{ SON } 0.051$ — winter heavier, consistent across three independent views, worth a deliberate discussion rather than a silent change.

“Why is the dependence trend not significant?” It is not yet tested: the constrained optimiser never left its starting point (identical objectives, exactly zero statistic), so the flat $\hat{a}(t)$

measures the optimiser. The model-free χ rises $0.25 \rightarrow 0.40$ and the unconstrained fit rises $0.40 \rightarrow 0.49$; the constrained refit with a smooth penalty is the outstanding fix.

“What is the strongest single sentence the marginal stage supports?” “At every series the high quantile and the overshoot scale are rising significantly, the threshold is calibrated locally by season and era, and the transform passes its uniformity gate in every slice — so the marginal intensification result stands on checks, not assumptions.”

Part IV — The fix list

1. **Dependence optimiser (blocking).** Smooth penalty in place of the wall; Nelder-Mead then BFGS; multiple starts including the unconstrained solution; print the violation at the solution. Until then, no dependence-stage number is quotable — including the functionals.
2. **Adopt the tensor threshold.** The local table settles it: first-era misses $0.149 / 0.123$ become $0.039 / 0.039$.
3. **Decide the conditioning question** with the professor: season-relative extremes (current) or flood-season extremes (freshet-only). The per-season table shows the choice changes the answer.
4. **Lift the $\tau = 0.01$ rung** (or re-tune the bottom of the ladder) so the below-rung share matches its nominal 1%.
5. **Raise the bootstrap** to $B = 2000$ with selections inside the loop, once the dependence fit is trustworthy.
6. **Bring in RCP 8.5**, turning the year contrast into the scenario contrast the display was rebuilt for.